Item 1 WAKISSHA

A friend of your family, working abroad bought a plot of land next to your home. He has built a hotel next to your home. He intends to install security gadgets such as cameras and others. He wants to secure good glass for windows for nice appearance as well. He doesn't know how to select good glass and security cameras. He has contracted an expert to help in searching for good glass and security cameras with standard of 1.50−1.70 range of reflective index for glass and focal length of about 10.0 cm − 15.0 cm range when light of speed $3.0 \times 10^8\,ms^{-1}$ is used.

Hint: Speed of light in samples of glass given and power of the lenses present are as shown below.

Glass	Speed of light	Power of the Lens from glass material
A	$1.97 \times 10^8\,ms^{-1}$	8.3D
B	$2.14 \times 10^8\,ms^{-1}$	5.0D

In addition, when the glass was covered with gelatin layer of green, and viewed from inside the house, he observed features outside the premises appearing so different than when viewed from outside; which excited him.

Task

As student of physics help the friend to;
(a) (i) understand the meaning of refractive index of a substance and give the causes of refraction.
(ii) identify which type of glass should be taken on for use.
(iii) identify which type of lens should be selected for use in security cameras.
(b) Understand the meaning of dispersion and explain why objects were appearing different when seen from the inside and when viewed from the outside.

ITEM 1 [JJEB]

Most people with eye defects prefer to regularly pay a visit to the Opticians for routine check-ups.

Jennie with her 15-year-old grandson, went to a shop that sells spectacles and ordered for a spectacle with lenses whose recommended power should not be less than **+8.29D** for her to see clearly. The spectacles were provided but Jennie wants to know if they can work for her. Her grandson was curious about the type of the lenses his grandmother bought, the eye defect she was suffering from and how the defect is corrected by the lenses purchased.

In the nearby test laboratory, monkeys were being quarantined in a sealed glass container one at a time and making a lot of noise. This was done by the doctors to determine if the monkeys were healthy such that they test the vaccines. The sound of a healthy monkey should not be of more than **1710Hz**. The doctor would turn on a certain pipe. Jennie and her grandson were left puzzled and asked the doctor to explain to them why the monkey's noise reduced gradually and suddenly disappeared when the pipe was turned on, even when the monkey's mouth was seen opening.

Jennie's grandson went for a short-call in the wash rooms of the laboratory and he could not understand how the curved mirror he was standing close to in the washroom produced an upright image smaller than his actual size. This type of mirrors has a number of applications in daily life situations. You have been contacted by the doctor and Jennie.

Hint:

When one of the lenses in the spectacles was placed 20 cm from an object, a clear sharp image was formed on a white screen 30 cm away from the lens.

Wavelength of the sound produced by the monkeys was 0.2 m.

Speed of sound in air is 340 ms^{-1}

Task:

(a) Help;

(i)Jennie and her grandson to address their concerns on the spectacles and explain to them why the monkey's noise disappeared. (include illustrations where necessary)

(ii) the doctor to find out if they can go ahead to test the vaccines.

(b) Explain any two daily applications of the mirror Jennie's grandson encountered in the washroom.

ITEM 1 [UMTA]

A town council installed a tall watchtower fitted with a high-powered searchlight to improve night security. During maintenance, a technician replaced the searchlight's original narrow-aperture concave mirror with a wide-aperture concave mirror of the same focal length. After the change, security guards observed that the beam failed to clearly illuminate the ground boundary areas.

As concerns about safety compliance for tall structures increased, regulations required that all buildings above 20 m in height must install a high-visibility warning light at the top of the structure. The town council was therefore required to select a suitable warning light. To verify whether the tower met the height requirement, the council requested an investigation.

S.4 Physics students from a nearby school were engaged to assist in the assessment. They first used a **20 cm** long pinhole camera placed **100 m** from the base of the tower and obtained an inverted image of the tower measuring **5 cm** on the screen. To verify their result, they also used an echo method. A sharp sound was produced at the base of the tower, and the echo reflected from the top of the tower was heard **0.14 s** later. The students were required to compare the results obtained from the two methods before advising the town council.

Hint: *Take the speed of sound in air as 340 m s^{-1}.*

Task:

(a) Help the tower owner understand how the change in the mirrors affected the reflected beam.

(b) Make a choice for the high visibility warning light. Justify your answer.

(c) Explain how the echo method can be used to determine the height of the tower.

(d) Determine the height of the tower using each method, compare the results obtained, and advise the town council whether the tower requires a high-visibility warning light.

ITEM 1 [KAMSSA]

A boy stood in a valley between two large cliffs separated by an unknown distance. He produces a loud shout and hears his own voice again after 4.0s and another after a further 1s. He also observes that sound is much clearer at night than during the day, which leaves him confused about the nature of the phenomenon they are experiencing.

In a separate observation, the boy is holding a glass block of refractive index 1.5.

A beam of light traveling in air is incident on the surface of this glass block. The student investigating the behavior of light also wishes to determine the conditions under which light would be completely reflected at the glass-air boundary. (Take speed of sound in air = 340 m/s)

Task

As a physics student:
(a) (i) Help the boy understand why he heard himself thrice.
(ii) Explain why sound is clearer at night than during the day.
(iii) Calculate the distance between the cliffs.
(b) Determine the critical angle for the glass-air boundary and state what happens when the angle of incidence in glass exceeds this value.
(c) Explain one application of the phenomena made in a separate observation.

ITEM 1 [UTEC]

During the enforcement of a Uganda trade order, an L.C.1 defense personnel was instructed to inform traders that street vendors and buildings located at least 0.025 km away were to be demolished. The personnel used a hired megaphone and a motorcycle equipped with a side mirror that continuously located distant objects as he moved. While on the tarmac road, the personnel observed that the surface appeared to be a glittering pool of water which kept disappearing as he approached it, which made him wonder greatly. While communicating, he bypassed a large workshop building and began hearing his own spoken words repeated at intervals of 0.175 seconds. The personnel stopped the megaphone using a remote control, but upon switching it back on and communicating, the sound repeated again and only cleared when he moved away from the structure. Consequently, he felt the megaphone had mechanical issues. The owner of the building heard the announcement, although he could not clearly see distant objects, and was in fear of the demolition of his structure.

Hint:

- Speed of sound in air is 320 ms^{-1}
- Ultraviolet radiation is characterized by high frequency.

Task:

Help:
a) Clear the personnel's wonder regarding the glittering pool of water on the road.
b) Using sketch ray diagrams, explain to the owner of the workshop his particular eye defect and its correction.
c) Explain the side effects of prolonged exposure to the rays used by the remote-control device that the personnel used to switch off the megaphone.
d) Inform the owner of the structure regarding whether his building meets the demolition criteria.
e) Using a clear ray diagram, explain why the motorcycle mirrors located those distant objects and discuss the challenges of using them.
f) Give more applications of such mirrors in daily life.

ITEM 1 [WENSSEC]

Physics students went on a tour where they observed several natural phenomena. On hot days, they noticed that the sky appears blue at midday but red in the morning. They also observed that

tarmacked roads, when viewed from a distance at an angle, appear to have pools of water; however, these "pools" disappear as one approaches them.

On another day, they experienced a sudden darkness at about 3:15 p.m. that lasted for approximately 7 (seven) minutes. Later, their teacher explained these observations in class using a magnifying optical instrument that produced a clear and enlarged image. One student who was seated near the instrument was unable to see the images clearly but clear while far away. The rest of the students were curious about what they saw

Frequency of Blue colour = 6.1 X10^{14} Hz

Frequency of red colour = 4.5 X10^{14} Hz

Speed of light = 3.0 × 10^{8} ms^{-1}

Tasks

Help the students to;
(a) Explain the above occurrences
(b) Describe with aid of diagram the optical instrument the teacher used for presentation
(c) Understand what happened with their friend and how it can be corrected

ITEM 1 [MASAKA]

During a science project, learners were required to make a solar cooker and a pinhole camera. The solar cooker they made was however not heating water. Agaba, a senior four student who came to help them made the following observations for improvement of the project;

- The part for heating the water had been placed at the radius of curvature.
- The inner part of the solar reflector was rough and also painted black.
- The reflector used was a normal concave mirror instead of a parabolic mirror.

None of the learners could make sense of the observations made by Agaba.

The students who were to make the pinhole camera wondered how long the pinhole camera had to be in order to produce an image of magnification 0.018. They also wanted to know how the camera would form images and why they were told not make the pin hole very big.
Hint; The distance between the object they wanted to view and where the screen of the pinhole camera is 20m

Use your knowledge of Physics to;

a) To help the students understand the observations made by Agaba.
b) To help the students establish the required length of the pinhole camera.
c) To help the students understand how the images would be formed and make sense of the advice given to them.

ITEM 2 [WAKISSHA]

Workers in a hospital are worried about the storage unit for a radioactive element used in treatment of cancer. They proposed to build a concrete bunker (barrier) in which to store it, but were not sure of it's dimensions. Upon consulting former employee of the Uganda Atomic energy council, he told them that radiation intensity, above 75 units is considered unsafe for humans.
The available data, indicating the relationship between thickness of concrete and the amount of the intensity that penetrates it is represented in the table below.

Intensity of radiations (units)	400	120	50	15	5.0
Thickness of concrete (cm)	0	10	20	30	40

Students who came to visit the hospital wondered why they needed to build a bunker and its exact dimensions. One of them even asked how these radioactive materials are used for detecting cancer in the body.

Task

Use your knowledge of physics to help the student understand;

(a) the safe thickness of the concrete.
(b) why the bunker has to be constructed.
(c) how the radiations are used to treat cancer.
(d) how nuclear energy is produced in a nuclear plant.

ITEM 2 [JJEB]

As one of the countries embracing sustainable development, Uganda started implementing "***Green energy***" policy by use of electric buses, and has started plans to generate more electrical energy using nuclear processes in Buyende district near lake Kyoga.

The ministry of energy and mineral development extracted and installed a radioactive mineral element in the nuclear reactor to start the process of production of electricity. At the time of installation of the mineral element, its count rate was 1000 counts per minute.

The "Green energy" experts advised the ministry that for the production process to be controllable, the production of electricity should start at the time when the count rate of the element is 10 counts per minute above the background radiation. The ministry is searching for a person to help it to know, how long it should take for the production process to start.

The people living in the surrounding community are puzzled and wish to understand how the too much water that the ministry plans to pump into the reactor, together with the mineral element help in the production of electricity.

They also want to be educated on the economic effects that come with the constructed power plant, and the health hazards associated with the power plant that made them get displaced.

Meanwhile, during the installation process, one of the workers was struck by a falling object which broke the bone of his arm. Two of the community members rushed him to the nearby hospital where his arm was subjected to a device that uses highly penetrative radiations that takes picture of fractures. The residents were curious to know how the device works and how it is used to locate the fracture in the worker's hand.

You have been approached.

Hint:

- ***The half-life of the radioactive element is 2 years.***
- ***The background radiation is 45 counts per minute.***

Task:

(a) Help
  (i) the ministry to establish the time electricity production will start.
  (ii) and address the concerns of the surrounding community.
(b) Explain to the two community members the principle of operation of the device and the application they wish to understand.

ITEM 2 [UMTA]

Frequent leakage has been detected in underground water pipes within a community. In attempts to locate the damaged sections, several pits have been dug along roads, leading to destruction of the road surface and disruption of transport. To address the problem, experts are considering alternative detection methods.

One group of Senior Four students suggests using a radioactive source that emits alpha particles to trace the leakage points underground. However, the experts are uncertain whether this method can effectively work in the given conditions.

A second group proposes using a radioactive tracer introduced into the water system with an initial activity of **1500** counts per minute, which has a half-life of 10 minutes. A Geiger-Miller (GM) tube is used to monitor radiation levels over time. However, the learners were worried about; how the tracer would be used, how long the water would take to be safe for use and required to know measures that needed to be taken to ensure safety.

Hint: Background radiation of the area is 50 counts per minute. Water is safe when its activity is approximately equal to the background.

Task:

(a) Help the students understand if the first method was reliable.

(b) Solve the concerns of the second set of learners.

ITEM 2 [KAMSSA]

Farmers in Kasese district grow vegetables near river Nyamwamba as a source of livelihood. The natives living near the riverbank have been experiencing health complications, which attracted the attention of the government through the Ministry of Health. The Ministry sent a nuclear physicist, to investigate the problem. After conducting a survey, the doctor reported that the vegetables contained radioactive materials. These radioactive substances were being deposited on the vegetables through contaminated water from the river, exposing the local population to harmful radiation. To determine how the radioactivity changes with time, samples of the contaminated vegetables were taken to a laboratory and their activity measured. The results obtained are shown in table 1.

Time (minutes)	0	10	20	30	40	50	60	70
Activity (Bq)	1600	1130	800	565	400	283	200	141

Table 1

The Ministry of Health considers the vegetables safe for consumption when the activity falls to 100Bq or below.

Task

As a Physics learner:

(a) State the radiations that may have affected the vegetables.
(b) Plot a graph of activity against time and use it to determine the time taken for the vegetables to be safe for consumption.
(c) State the dangers of exposure to these radiations and possible safety precautions that the natives should take against radioactive contamination.

ITEM 2 [UTEC]

A nearby mine in a certain region of Uganda was suspected of discharging dangerous radioactive materials into the nearby swamp water that supports community activities. A study was carried out, and the results are presented below:

Time (hours)	0	10	30	46	60	90	118
Count rate (per hour)	625	575	500	450	400	300	250

The study also showed that the swamp should remain out of reach to the community until the count rate drops to 220 counts per hour; this finding was not received in good faith by the community. It was further suggested that the whole community should undergo individual medical check-ups if the half-life of the substance is above four (4) days.

The community also alleged that the mine consumes too much electricity, which affected their local supply. However, the miners defended themselves by stating that they have a plan to produce more reliable electricity through the nuclear fission of uranium, as represented by the following equation, though the community still doubted it:

$$^{235}_{92}U + {}^{1}_{0}n \longrightarrow {}^{144}_{x}Ba + {}^{y}_{36}Kr + 3{}^{1}_{0}n + energy$$

Task:

a) Help the community members to:
(i) Determine the estimated number of days the community members need to keep away.
(ii) Know whether they will be subjected to further individual medical tests.
b) Kindly inform the community members to help them appreciate the resolution to stay away for the determined duration.
c) Using a well-labeled diagram, explain the possibility of producing electricity from the nuclear fission process described above.
d) Present a balanced nuclear equation by finding the values of x and y.
e) Explain the uses and international regulations regarding the use and control of such radioactive materials today.

ITEM 2 [WENSSEC]

A nuclear laboratory was recently established in a city. Following an earthquake, radioactive material with an activity of 750 counts per minute, as measured using a Geiger- Müller (GM) tube, spilled onto the laboratory floor. After inspection by nuclear scientists, all workers were

instructed to evacuate the area immediately, although no specific time for re-entry was given. To enhance security, an x-ray screening system was installed at the laboratory entrance to inspect people and items entering or leaving the facility.

The laboratory owner is seeking clarification on the following issues:

(a) Why the workers were instructed to leave the laboratory

(b) How long the laboratory will remain closed before it is safe to re-enter

(c) How the x-ray screening system was used to improve laboratory security

Background radiation in the area = 50 counts per minute.
Half-life of the material =2 months

Tasks

Help the laboratory owner to;
(a) Address the above challenges
(b) Describe one benefit of radioactive material to the community

ITEM 2 [MASAKA]

Due to an increase in pests in a certain garden, an agronomist has suggested a pesticide that has radioactive materials with an activity 1200 counts per minute which reduced to 37.5 counts per minute after 15 days to be sprayed on the crops.

As soon as the community members got to know this, there have been increased threats from the wholesalers who buy produce from the farm to boycott their produce since they feared the vegetables were dangerous to eat and they were not sure if the right pesticides were used.

The workers at the farm however did not know why there was a worry, and even then, they could not tell how long it would take for the vegetables to be safe for consumption. They consulted a scientist who told them that in as much as the radioactive materials could be useful, there was reason to worry. He however did not have the time to answer their other concerns.

This left them wondering how the materials were useful in life.

Hint: Radioactive materials suitable for this purpose should have a half-life less than 4 days.

Background radiation count in that area is 55 counts per minute.

Use your knowledge of Physics to:

a) Determine if the pesticide was suitable
b) Ascertain if the vegetables were safe to eat 15 days after spraying.
c) Help the farm workers understand why the wholesalers were worried.
d) Briefly describe how radioactive materials are useful.

ITEM 3 [WAKISSHA]

A son who stays in South Africa's Cape Town invited his father for a visit. The son's home is near the ocean and on one evening the father saw the water level raising beyond the normal but

later on went back. The old man got worried. While at his son's home, they were able to watch Uganda channels on television and the man wondered how signals could travel that far. One day he wanted to go to the mall for shopping, there was no body to escort him. But the son told him to use G.P.S to locate the mall and then back. He also wanted to know how it works.

Task

Using your knowledge of physics,

(a) Help the old man understand;
  (i) why water was raising beyond normal?
  (ii) the working of the G.P.S system.
(b) Explain to the old man some of the benefits of the raising water levels he observed.
(c) Describe how live broadcast is possible by television across different countries.

ITEM 3 [KAMSSA]

The government of Uganda, in collaboration with an international space research organization, has launched a lunar exploration mission called MoonLink- 1. The mission aims to study the Moon and establish a communication system between astronauts on the Moon and scientists on Earth. During the mission, students from a Secondary School were invited to follow the progress of the spacecraft. They observed that the Moon appeared to change shape over several nights. The students also wondered how astronauts on the Moon could communicate with people on Earth despite the large distance involved.

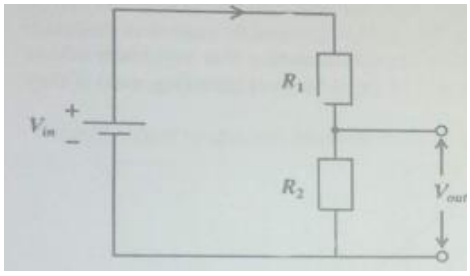
To ensure reliable communication, the spacecraft uses an electronic control circuit containing logic gates and a potential divider network. The communication satellite is placed in orbit around the Moon to relay signals between astronauts and Earth.

The communication system activates an emergency transmitter only when:

- Signal detector (A)=1 and
- Power monitor (B)=1

The emergency transmitter output is represented by Y

A potential divider consists of two resistors connected across a 12V supply such that $R_1 = 4k\Omega$ and $R_2 = 2k\Omega$. The output voltage is taken across R_2 as shown below:



The potential divider can only work if the output voltage across R_2 is greater than 3V.

Task

As a learner of physics,

a) Explain why the moon appears to have different shapes at different times of the month.

b) (i) Explain how communication between astronauts on the Moon and scientists on Earth is possible.
(ii) State two advantages of using satellites in space communication.
c) The emergency transmitter operates according to the logic condition: $Y=A\cdot B$
Draw the logic symbol representing the circuit and its truth table.
d) Determine whether the potential divider can work.

ITEM 3 [UTEC]

In February 2026, the USA and Israel engaged in a military conflict with Iran in the Middle East, during which Iran's Supreme Leader was assassinated at around 9:30 AM local time in Iran. The attack was closely monitored by the White House in the USA using space-based satellites, which asserted that the incident occurred at around midnight USA Standard Time. The Islamic Republic of Iran retaliated by launching sophisticated missiles at the State of Israel that penetrated the Israeli defense system, since the missiles traveled for almost 9 minutes at a hypersonic speed of Mach 13. As the war escalated, the entire planet Earth within the Solar System was affected, particularly as fuel prices skyrocketed.

Support Material:

- One Mach speed is equal to the speed of sound in air.
- Speed of sound in air is 330 ms^{-1}.

Task:

a) Explain the cause for the difference in the recorded time of occurrence of the attack that claimed the life of the Supreme Leader.
b) How was the White House able to observe and follow the attacks in such a distant country?
c) Estimate the distance between Iran and Israel based on the missile flight data.
d) Discuss other components of the Solar System.
e) Give several other applications of the satellite device setup that enabled the White House to follow up on the attack.

ITEM 3 [UMTA]

Fishermen in Kalangala on Lake Victoria face serious challenges during night fishing. Thick mist and storms make it difficult to find direction, while regular changing water levels sometimes make landing and docking unsafe.

To improve safety, the Government invited experienced fishermen from Canada to share technologies used in their fishing industry. The visitors explained that in July, Canada experiences much longer days and shorter nights than Uganda, which reduces the time available for night fishing. This discussion led to questions about the Sun and other stars. One fisherman asked how the Sun produces enormous amounts of energy for millions of years and whether stars become hotter or cooler as they grow older.

The visitors also demonstrated a boat safety system that uses a bistable control circuit to automatically control the boat's engine and a GPS system to locate boats during bad weather. Although the fishermen appreciated the technology, they wanted to understand the science behind it after the Canadian team left.

Task:

(a) Explain;
(i) Fishermen's challenge on the water levels.
(ii) Visitor's explanation.
(iii) Fisherman's question.
(b) Explain how the GPS satellite system can be used by the fishermen.
(c) Design a bistable switch circuit using two NOR gates for the automatic control of engines and construct its truth table.

ITEM 3 [WENSSEC]

A tourist visited a coastal town and observed several phenomena that surprised him. He checked into his hotel room at about 9:00 a.m. when the coastline was approximately 200 m away. However, after about two hours, he noticed that the coastline appeared much closer, about 10 m from the position where he had originally stood, and he wondered whether this occurrence had any benefit to the community. At night, he observed a sky filled with many stars of different colours and brightness and they twinkling, he wondered how they came into existence and why they appeared in a such manner. He also noticed that the appearance of the moon kept changing over time. In addition, a security alarm system using a photocell was installed at this entrance door to prevent unauthorized access to his room, which further puzzled him.

Later, when he attempted to contact his relatives using a telephone, his call was successfully connected. He was surprised, as he believed telephone calls could only be made within the same country and through cable networks.

Tasks

Help the tourist to understand;

a) Why the coast line kept shifting and how it is beneficial to the community
b) How the stars came into existence and why they appeared twinkling, with different colours and bright
c) Why the appearance of the moon kept on changing
d) How a photo-based alarm system operates
e) How relatives were able to receive the call

ITEM 3 [MASAKA]

Salim, a horticultural farmer recently went to another country. Upon return, he was surprised by the difference in seasons between the two countries. Due to the extremely high temperatures, he decided to set up a greenhouse with an automatic temperature and humidity control system that turns on a fan with a sprinkler whenever temperature exceeds 30∘C or humidity falls below 30% using an appropriate logic gate system. No one could explain to him how this system could be set up.

When Salim went to electronics shop to buy bulbs, he was given Light Emitting Diode (LED) bulbs instead of filament ones. The person at the shop could not explain his choice.

Use your knowledge of Physics to;

a) Help Salim understand the cause of the difference in seasons.
b) Explain to Salim how the system in the green house can be set up and identify other areas where it can be used.
c) Help the worker give reasons for his choice.

ITEM 3 JJEB

On 1st April 2026, NASA sent a group of Astronauts (Artemis II crew) in a lunar flyby mission that lasted up to 10th April 2026. One of the pictures taken by the crew was used by a teacher in one of the secondary schools in Uganda for a lesson. The learners wanted to know why the earth had one part with light and the other side dark.

Support diagram

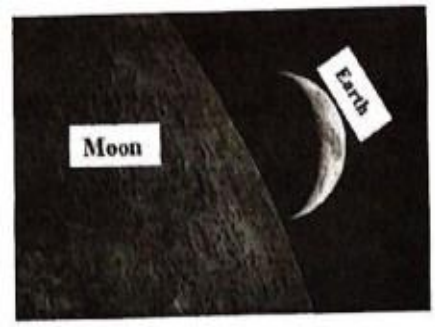


Photo taken by the crew on one side of the moon

A tourist from a winter-stricken nation flew to a certain island on 3rd April 2026 for a vacation. Upon his arrival, he noticed that the island was experiencing a very dry summer climate, despite being completely surrounded by an ocean. He could not understand the difference in seasons of his home country and the ***island.***

Furthermore, he also noticed a strange phenomenon, where in the morning hours, the ocean's water level was very low, but by the evening, the water level becomes very high. On 9th April, the tourist prepared to visit Mpoma Satellite station in Uganda to clearly watch Artemis II flyback from the mission, but he had no idea how to locate the satellite station.

You were contacted for help, and you sent the tourist a navigation app ***(Citymapper***) to install on his smart phone. The tourist watching the crew moving in the earth's atmosphere, was puzzled why the Capsule outer surface was emitting fire.

Task:

(a)Help the:

(i) students understand their observation.

(ii) tourist address his concerns when on the island.

(b) Explain to the tourist how the app can help him and explain why it is important for a developing nation like Uganda to invest in space exploration.

ITEM 4 [WAKISSHA]

A group of scouts on a camping trip noticed something peculiar about their flag at the lake shore. During hot days the flag flew towards land and at night, it flew towards sea even when there was no wind. They were puzzled by this observation. Later they met a local farmer who had recently built a dam on his farm. He was too unsure whether to fill the dam to its full capacity and why the walls of the dam were so thick at the bottom.

Additionally, the farmer was concerned about the quality of milk produced on his farm and suspected that his employees might be diluting it.

Hint*: the dam is 8 metres deep and can support a maximum pressure of 70,000 Pa. Density of water = 1000kgm^{-3}*

Task

As a student of physics;

(a) explain to the scouts why the flag behaved that way.

(b) help the farmer understand the purpose of thick walls at the bottom in the dam and advise him on filling the dam to capacity.

(c) suggest a way for the farmer to determine if the milk was diluted or if there were other factors affecting it.

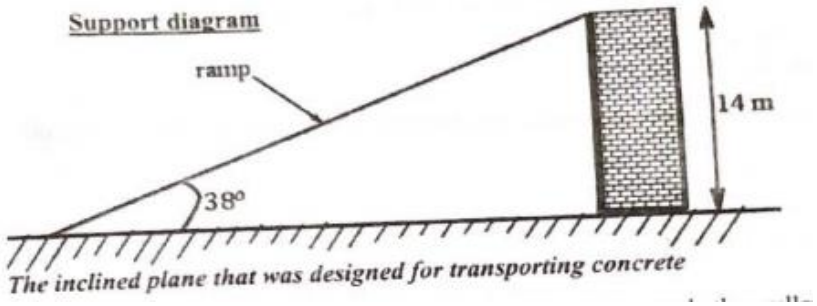
(d) state the applications of the atmospheric pressure.

ITEM 4 [JJEB]

During the construction of a storeyed building near the lake shore, workers are made to lift the concrete of mass **250kg** using an inclined plane and pulley system containing a pair of pulleys in each of the upper and lower blocks.

The Chief Engineer's Requests are:

(i) how the pulley system should be arranged and workers should use the system whose efficiency is not less than **71%**.



The inclined plane that was designed for transporting concrete

The inclined plane that was designed for transporting concrete

(ii) Joby monitoring the pulley system should always reach the pulley area at a velocity not less than **4.3ms $^{-1}$.** The engineer wants to know if **Joby** will manage the task or whether an automated machine system should be installed to monitor the pulleys.

Joby upon reaching the pulley area, has to remove the lower hot pulley whose temperature is **110 ∘C** and insert it quickly into **231g** of ice blocks placed in a metallic container such that the pulley cools after which he fixes the pulley back into its place if the temperature becomes not more than 37.8 ∘C. He wishes to know whether to reduce or increase the ice blocks used.

Joby could not understand why;

(i) he became tired while climbing to remove the pulley but did not feel the same way when lowering the pulley into the ice blocks.

(ii) why he felt cool air blowing towards him when he was taking a walk along the lake shore.

Joby's *work-mate* was curious about the principle used in cooling the pulley and wants to know why water instead of cooking oil is used in car radiators.

You have been contacted for aid.

Hint:
Job's workplace is **20 m** away from the pulley area and takes him 10 seconds to reach the pulley area.
Specific latent heat of fusion of ice is $\boldsymbol{3.36\times10^{5}J\ kg^{-1}}$
The container in which the Ice blocks are placed weighs ***3.0 kg***
Specific heat capacity of the container in which the ice is placed is $\boldsymbol{2100\ J\ kg^{-1}K^{-1}}$
Force applied on each of the systems is ***3750N***
Mass of each pulley in the lower block is ***10 kg***.
Specific heat capacity of water is $\boldsymbol{4200\ Jkg^{-1}K^{-1}}$
Specific heat capacity of the material of the pulley is $\boldsymbol{490\ J\ kg^{-1}K^{-1}}$
Acceleration due to gravity is $\boldsymbol{10\ m\ s^{-2}}$

Task:

(a) Read the information in the scenario and help
  (i) to address the engineer's requests.
  (ii) **Joby** to solve his problems.
(b) Explain to Joby's work-mate how the principle is applied to the car radiators.

ITEM 4 [KAMSSA]

A parent in a certain village recently gave birth to a baby. The doctor advised that the baby should be bathed using water at a temperature between 29°C and 35°C to avoid skin burns. To prepare the bath water, 2 litres of hot water at 70°C is mixed with 4 litres of cold water from an underground well at 15∘C in a metallic basin. Before the baby was bathed, the child's father was called away to welcome a visitor, leaving the water exposed to the environment. Upon his return, he decided to pump more cold water at 15∘C from a nearby well using a force pump before mixing it with the hot water.

Support
Specific heat capacity of water = $4200Jkg^{-1}K^{-1}$
Specific heat capacity of basin = $900Jkg^{-1}K^{-1}$
Mass of metallic basin = 6kg
Density of water = $1000\ kgm^{-3}$
1 litre of water = $10^{-3}\ m^{3}$

Task

As a physics learner;
(a) Determine whether the final temperature of the water mixture is suitable for the baby.
(b) Advise the family on ways of keeping the hot water warm while they are away welcoming the visitor.
(c) Explain how a force pump works to raise water from the well and deliver it to a higher level.

ITEM 4 [UMTA]

A company designing a new playground seesaw wants to improve safety and comfort by attaching impact-absorbing springs at both ends. To ensure the springs are strong enough, safety guidelines state that the material must have a Young's modulus of at least $\mathbf{3.11\times10^{5}\ N\ m^{-2}}$. To

verify this, a sample spring with an original length of **15.0 cm** and a cross-sectional area of **20 cm²** is tested in a laboratory, producing a load-extension graph shown in Fig. 1.
While assisting with the assembly of the first model, a casual labourer notices that the main frame consists of a hollow pipe instead of a solid rod. He wants to know the exact mass of the pipe to write down in his daily record book, but the paper showing the weights of the materials is completely torn. He decides to find its mass experimentally using basic tools available at the workshop but has no idea how to set up the test.
Furthermore, while testing the seesaw by moving it up and down, he is surprised to notice that a significant amount of heat is generated at the pivot. He comes to you for explanations and guidance regarding:

(i) Assessing whether spring meets safety requirements based on its Young's Modulus.
(ii) Explaining how the mass of the hollow pipe can be found using the principle of moments.
(iii) Justifying the use of a hollow pipe instead of a solid one in the above aspect.
(iv) Explaining why heat is produced at the pivot.

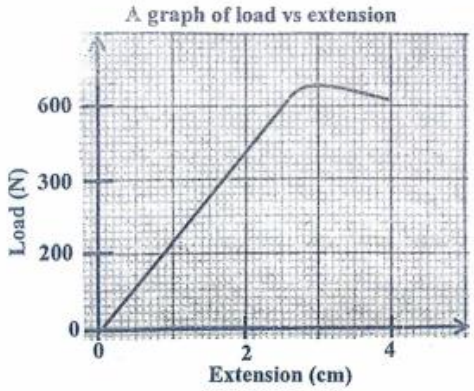


Fig.1

Hint: The spring extends to 17.5 cm under maximum load.

Task:

Use your knowledge of Physics, avail the labourer with the information that he requires.

ITEM 4 [UTEC]

John hired a special car to travel to school, which charged Ughs 5,000 per kilometer on board. While loading luggage, John placed his bags on the top rack, but the driver changed the arrangement and put them in the lower cabin, which John did not like. As they started the journey, John hit his head hard against the seat when the car suddenly moved. He alarmed the driver, who stopped the car immediately, causing John to hit the forward seat. The driver then started slowly from rest and accelerated at a rate of 0.1 m/s^2 for 4 minutes. The driver continued at this steady speed for 10 minutes and bypassed an automatic penalty billing machine that fines every driver who exceeds a speed of 90 km/h for over speeding, and then decelerated uniformly to arrive at the school in 3 minutes. The candidate offboarded and was charged him beyond the agreed tariff. The candidate also developed a severe headache; when taken to the sickbay, the nurse attributed it to either fatigue or malaria. She checked his temperature, which read **310.6.** the nurse required that they pay for a further malaria test if the temperature exceeded Upon arrival, a health worker checked John's temperature and found it to be 37.8 $^{\circ}C$,

Task:

Help:

a) (i) Explain the physical effect experienced by John when the car started and stopped, and how it could be limited.
(ii) Give applications of this effect in daily life.
b) Inform the driver whether he was billed for over speeding by the automated machine.
c) Help John appreciate the driver's decision regarding the position of the luggage.
d) Determine whether the candidate was required to pay for the further malaria test based on his temperature.
e) With reference to a sketch velocity-time graph, determine if the fare charged by the driver was ideal based on the distance covered.

ITEM 4 [WENSSEC]

A number of issues arose at a construction site after the laying of a concrete slab. One of the supervisors of a construction site wearing sharp-pointed shoes found it difficult to walk on the newly laid wet surface compared to other workers wearing flat-soled shoes.
The builder suggested that the concrete slab on the first floor should be reinforce, unlike the one on the ground floor. However, the site owner did not agree, and the builder was unable to give a clear explanation for his proposal. In addition, lifting concrete blocks to the first floor became difficult and tiresome. The workers reported that the height of 6 m was too high for them to safely throw the blocks upward. Some workers who attempted to do so ended up breaking the blocks when they hit the ground, producing a loud sound.

In response to these challenges, an engineer proposed the use of a pulley system with three wheels connected to a platform carrying two concrete blocks, each of mass 6 kg. The engineer was concerned that the efficiency of the pulley system might be less than 100% and was uncertain whether it would effectively reduce the workers' effort.

- Both supervisors have a mass of 70kg
- Acceleration due to gravity is $10ms^{-1}$
- 18% of the work input is wasted in the operation of the pulley system.

Tasks

Help the supervisor to understand:
(a) Why it was hard for one of them to walk on the newly laid wet concrete surface.
(b) The benefit of the builder's proposal of reinforcing the concrete on the first floor.
(c) Why the blocks made noise while hitting the ground
(d) Whether the pulley system eased their work.

ITEM 4 [MASAKA]

In a certain school, a pump of power 3500W that is meant to draw water from a filled underground tank of volume 30000 liters at a depth of 15 m recently broke down after working for 8 minutes. They however couldn't tell how much water had been drawn by that time.
One of the learners suggest they adopt the use of a lift pump but others argued it could not and suggested the use of a force pump instead.

The other learners wondered how the force pump would be used to draw water and were surprised to see that the metallic part of a hoe left outside by the workers through the night felt colder than the wooden one when touched.

Use your knowledge to;

a) Determine how much water was drawn from the tank.

b) Give a rationale for why the lift pump was not suitable for that purpose.
c) Help the learners understand how the alternative pump suggested would solve the problem.
d) Explain why the materials felt different when touched.

Item 5 [WAKISSHA]

In mountainous region, most people stay at the foot of the mountains while others stay at the top. People at the top experience water shortage and nose bleeding while those at the foot experience severe heat during dry spell. An NGO, a voluntary organization wants to help the people reduce on water shortage by installing a solar pump to provide constant water supply of 300 litres every minute but they do not know the height of the mountain. They have contacted a meteorologist who gave them a barometer to help them in this task.
Hint: Barometer gave that pressure difference between the top and bottom as 10 kpa.

- Density of air is 1.25kgm $^{-3}$
- Acceleration due to gravity is 10 ms^{-2}

Task

As a physics student, help the NGO and settlers to:
(a) explain why it is hotter at the bottom and cooler at the top.
(b) explain the causes of nose bleeding and water shortage at the top of the mountain.
(c) estimate the height of the mountain.
(d) determine the power of the pump required to raise water to the top of the mountain.

ITEM 5 [JJEB]

Families, especially the working class, prefer shopping on weekends what is required for the following weekdays. Nelkon and his daughter, Jalia, went to a storeyed shop to buy some goods. They observed the following phenomena:

(i) Nelkon, watching a movie on the shop TV screen, saw a gun moving backwards as the bullets were being fired out of the gun barrel.
(ii) After buying a balloon and inflating it, the balloon accidentally moved out of Jalia's hands and when the mouth of the balloon opened suddenly, it rushed upwards as the air moved out in the opposite direction. *These left both Nelkon and his daughter puzzled.*

At the shop, there was giving prizes to a person who throws a wooden ball vertically upwards to reach a height of **10 m** above the ground level. Jalia, who is **140 cm** tall, was a participant. She threw the ball which took **1.2 seconds** to reach its maximum point. She wants to know if she won the prize and if not, what does she have to do next time?

Jalia's teacher requested her to boil a metallic steel ball in water to **100 °C** and pour the water together with the ball in the flask and cover it. The following day Jalia will use the steel ball, which will be still at **100 °C,** to explain to the class during the lesson, how the ball's specific heat capacity is obtained through a scientific investigation. Jalia is concerned about the possibility of the steel ball remaining hot up to the lesson time, she also doesn't know how to carry out the scientific investigation requested by her teacher.
You have been approached by Nelkon and his daughter, Jalia.

Hint:

Acceleration due to gravity is 10 ms^{-2}

Task:

(a) Your work is to:

(i) Explain to Nelkon and his daughter, Jalia to understand the physics principles behind their observations.

(ii) Find out if Jalia won the prize and if not, what does she have to do to win it next time.

(b) Help Jalia:

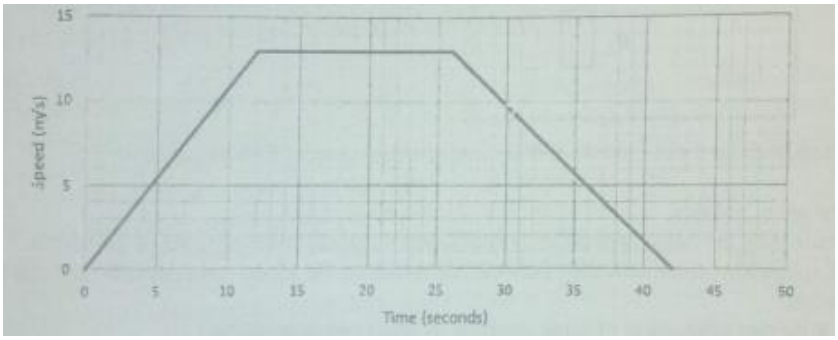
(i) understand the possibility of the steel ball remaining hot up to lesson time.

(ii) to respond to the teacher's request in class.

ITEM 5 [KAMSSA]

A driving school vehicle is being tested on a straight road between two traffic lights as part of a road safety assessment. The vehicle starts at the first traffic light and its motion is recorded. During the test, the driving school has a strict safety regulation stating that any driver whose average speed exceeds 10ms $^{-1}$ during the journey will be fined for over speeding, even if they obey traffic lights.

The velocity-time graph below shows the journey of a car between two sets of traffic lights.



The instructor later asks learners to explain the motion of the vehicle using the velocity-time graph provided, determine whether the driver should be penalized and explain the importance of seat belts.

Task

As a physics learner, Respond to the demands of the instructor

ITEM 5 [UMTA]

A restaurant that uses electric cookers wants to automate its food-warming system so that meals are always served within a suitable temperature range while minimizing electricity consumption. To achieve this, the manager proposes installing a thermostat fitted with a bimetallic strip to regulate the heating circuit automatically.

While restocking the kitchen, the chef recommends purchasing cooking pots designed with a low heat capacity and thick insulated handles instead of cheaper, similar-looking pots with a higher heat capacity. After installation, the chef notices that some pots are unstable when partially filled and decides to investigate their stability by determining their centre of gravity experimentally before use.

To test the performance and stability of one of these newly purchased aluminium pots, the chef uses it to prepare the daily offal sauce. To keep the food fresh before cooking, **1.0 kg** portions of the sauce are stored and frozen at **-20°C**. Before serving, each portion must be heated to **90°C** inside the saucepan. The restaurant owner has allocated Sh**. 3,500** to cover the fuel energy cost for heating one portion of the frozen sauce and wants to know whether this budget is sufficient.

Support Material

- Mass of one aluminium pot = 0.20 kg
- Specific heat capacity of aluminium saucepan = 900 J kg^{-1} K^{-1}
- Specific heat capacity of frozen sauce = 2,100 J kg^{-1} K^{-1}
- Specific latent heat of fusion of the frozen sauce = 3.34×10^5 J kg^{-1}
- Specific heat capacity of liquid sauce = 4,200 J kg^{-1} K^{-1}
- Cost of electricity: Every 50,000 J of heat energy consumed costs Shs. 200

Task:

(a) To help the restaurant owner understand how the proposed system of using a thermostat will solve the problem at hand and identify other instances where it can be used.
(b) Describe how the chef can carry out his investigation on the center of gravity.
(c) Explain why the chef recommended a pot with low heat capacity for the restaurant in terms of efficiency and safety.
(d) Determine the total electrical energy required for the complete heating process of one portion and state whether the allocated budget is adequate.

ITEM 5 [UTEC]

Senior 1 learners set off for a study tour to the source of a certain river. While traveling in a saloon car to the site, a speeding Sinotruk bypassed them, and they experienced a force pulling them towards the truck, which made them fear greatly for their lives. On reaching the site, two girls, A and B, rushed to a seesaw where Girl A, of mass 46 kg, was seated at 2.5 m from the center pivot. Girl B, who weighed 600 N, wanted to balance the seesaw on the other end but did not know the exact distance where she should sit. They then witnessed some ice floating on water and also noticed that the river was flowing fastest at narrow sections compared to wider sections, which puzzled them. They continued to a dam where their teacher cautioned them that swimming in a dam is only safe if the depth does not exceed 4 m. The tour guide informed them that the dam had thicker walls at the bottom to counteract a hydrostatic pressure of around 6.5×10^4 N/m^2, and advised that they should not risk moving near the water because the dam appeared shallower than its actual depth.

Hint:

- Density of water is = 1000 kgm^{-3}
- Acceleration due to gravity is 10 ms^{-2}

Task

Help

a) Explain to the learners the cause of the force that caused their fear while on board next to the moving truck.
b) Determine the distance Girl B should sit from the pivot to balance the seesaw, and state the conditions necessary for it to remain in this equilibrium state.
c) Explain the anomalous behavior of ice to the girls and explain any of its applications in daily life.
d) Inform them whether they will be safely allowed to swim based on the maximum depth rule and the given pressure threshold.
e) Explain any other applications of the knowledge of fluid pressure acting normally on surfaces (such as the bottom of a dam) in daily life.

ITEM 5 [WENSSEC]

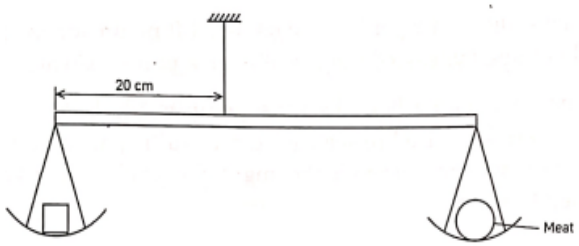
Maria is preparing food for a sick patient admitted at Arua Referral Hospital. She heats 2kg of cooking oil in a 0.5kg aluminum saucepan from 25∘C to 180 ∘C using a charcoal stove. Maria decides that she will only take the oil to the hospital if the total heat energy supplied reaches 0.7MJ. During the heating process, she notices that the cooking oil cools faster than water had done on a previous occasion and wonders why this happens but her friend advised her to keep the hot liquid hot using a vacuum flask, this puzzled her more.

Tasks

ITEM 5 [MASAKA]

A certain mixed farm is required to weigh their vegetables before selling them off. Customers' complaints have increased over the last week after acquisition of a new weighing scale.

When consulted on the weighing scale they use, he said a 1kg mass is put one side of a uniform wooden beam of length 50 cm and mass 100 g pivoted at the 20 cm mark. The weighing scale appears as it does below.



One of the clients wondered if the scale was fair and if a metallic beam of the same mass wouldn't serve a better purpose than the current one.

The farm needed to preserve the milk collected during that season. The manager procured a refrigerator. He wondered how it would be able to solve their problem since the only visible part he could recognize were thin black fins at its back. He wondered why they were made in that way.

Use your knowledge of Physics to;

a) Help solve the client's confusion.

b) Help the manager understand how the procured item would solve their problem.

c) Explain to the manager why the fins look the way they do.

Item 6 [WAKISSHA]

The district chairman is spear heading a project to start power generation from a certain water fall in the district to increase the supply, within the area. The project intends to produce 50,000 W at 500 V of a.c to be transmitted over a distance of 10km across the district. The total resistance of the transmission cables that have been connected on the poles for transmission is 4Ω per km. However, the house hold appliances in the region requires a main supply of 240V for proper use without any damage. The expert contracted also intends to sensitise the people on the recommended format and correct wiring system that will safeguard the devices connected on the grid.

Task

Using your knowledge of physics:

(e) help the chairman understand the advantage of generating A.C instead of D.C and guide him on the colour codes used in domestic wiring.
(f) explain to the chairman the working of the device that must be installed, through which the power must pass before it is connected for home use.
(g) determine the percentage power loss that occurs during the transmission.
(h) advise the chairman on ways how power loss during transmission can be reduced.

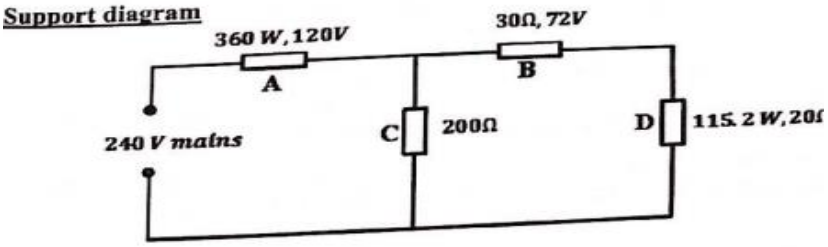
Item 6 [JJEB]

A house wiring system requires careful consideration of the wires used and the appliances to be connected into the sockets such that no electrical damage occurs.

Johnson who has just rented a house has been told by his landlord that he has to pay **1655 US dollars** as a fee for electricity per **30 days.**

Johnson claimed that the amount quoted was a lot and requested the landlord to refund his rent such that he finds another house. The landlord counter-claimed that Johnson had appliances of higher power ratings that would command the quoted amount but accepted to refund Johnson the rent if payment is found to be lower than what he requested Johnson to pay.

The landlord pulled out the connection plan showing the arrangement of Johnson's appliances A, B, C and D which are shown in the support diagram below.



Johnson's son, who is in S.3, told the father that the house wiring cannot help the appliances to work properly and others would easily burn. This angered the landlord who told Johnson that for the appliances to work properly, the former tenant in the same house had installed a device across each appliance to work at full voltage of the mains, should the current from the source be less than 3.5A.

Johnson's son also complained that their house was not protected from lightning and requested the landlord to apply some knowledge of electricity to protect their rented house in case they don't leave the place.

You have been contacted by the landlord.

Hint

- Cost of each unit of electricity is **9.6 US dollars.**
- Every appliance works for **8 hours** per day for a duration of **30 days**.

Task:

Your work is to address the following engineering and financial queries:

(a) (i) Find out whether Johnson has to be refunded the rent, explain to him why the appliances are most likely to burn and how to minimize the effect.

(ii) Determine if Johnson needs to install the device suggested by the landlord and explain how it can help to make the appliances work properly.

(b) Explain how the landlord can apply the knowledge of electricity to protect his tenants and their properties as claimed by Johnson's son in the scenario.

Item 6 [KAMSSA]

Your uncle owns a maize milling business and uses an electric milling machine to grind maize into flour. One day, the milling machine suddenly failed to start even though it was connected to a 12V power supply. He took the machine to a technician, who informed him that the relay switch controlling the motor had become faulty and needed replacement. The technician explained that the relay switch is designed to automatically switch on the motor when the current in the relay coil exceeds 1.2A. To test the relay, a coil of resistance 8Ω was connected across the 12V supply.

However, your uncle could not understand how the relay switch works or whether the milling machine would actually switch on under those conditions. He has therefore approached you, as a Physics learner, for assistance.

Task

Provide the information needed by your uncle.

ITEM 6 [UMTA]

A school has installed a new fire alarm system in its dormitories to protect students from fire hazards. This safety system uses an **NTC** thermistor as a temperature sensor and an **NPN** transistor as a switch to control an electric bell alarm.

Inside the system's circuit, a **15 Ω** resistor and the **NTC** thermistor are connected in series across a **12V** battery to form a potential divider as shown in **Fig. 2** below. This potential divider regulates the base voltage of the transistor, while the collector side of the transistor is connected directly to the electric bell.

Under normal room conditions, the thermistor maintains a high resistance of about **800 Ω** which keeps the transistor turned off. However, when a fire breaks out and the temperature rises, the thermistor's resistance drops down to about **50 Ω**. This sudden drop changes the voltage across the potential divider and turns the transistor switch on. The electric bell then rings to alert the students, but it will only sound once the current flowing through it reaches at least **0.15 A.**

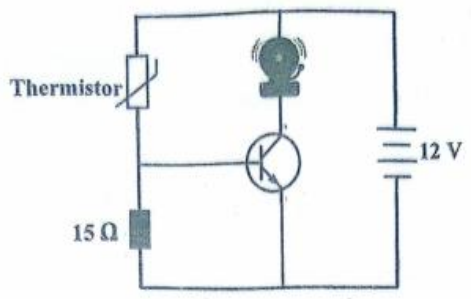


Fig: 2. *The fire alarm system circuit.*

Task:

(a) Explain how the thermistor and transistor switch will control the alarm system.

(b) Explain how the electric bell works and use the given data to determine whether it will sound at high temperatures.

Item 6 [UTEC]

A poultry farmer in Mityana is required to connect four brooder infrared bulbs, each labeled 250W, 240V, to warm chicks, and a security bulb to be used on the compound. These are connected to the mains electricity supply that charges Ugsh 760 per unit, with a fixed service fee of Ug.sh 3,500 per week. He has two switches and a junction box to connect the bulbs, as well as a socket for a speaker used for advertising. The farmer needs to limit electricity consumption by minimizing the hours of use of the security bulb and the brooder bulbs as much as possible since they are highly energy-consuming. The farmer has been using a local heating charcoal stove at a cost of Ugsh 65,000 per bag of charcoal per week, and is not sure whether he will be saving money by switching to the brooder bulbs. He plans to use the brooder bulbs for 10 hours daily and strictly use the security light during the hours of darkness daily (assumed to be 12 hours).

Task:

As a learner of physics, help the farmer to:

a) Make an informed decision between using the infrared bulbs and the charcoal stove by comparing costs.
b) Make a safe and efficient circuit diagram layout for the above appliances in reference to costs, safety, and efficiency.
c) Explain how the advertising speaker device works.
d) Automatically switch on and off a compound bulb effectively in response to darkness.
e) Give more applications of the rays used by brooder bulbs in daily life.

ITEM 6 [WENSSEC]

In Arua city, many households rely on electricity but often face power outages and high electricity bills because of unsafe wiring and lack of knowledge on how electricity works.
The residents have noticed that the connecting wires get very hot when operating on electricity and wonder why this happens. They also complain that their devices often get burnt because the electricity from power lines comes at high voltage and must be lowered before it reaches homes.

Many families struggle to understand their monthly electricity bills and how different wiring parts in their homes help protect them and keep the power flowing safely.

Hint:

Your uncle has a fridge rated (7.5A, 240V), eight light bulbs each rated (4A, 5Ω) and three smart phones each rated (4500mA, 240V).
Each unit of electricity is equivalent to 1 kWh.

Task:

As a learner of physics, your work is to,

(a) (i) Explain to the villagers why connecting wires become hot when operating
(ii) Help your uncle understand the meaning of the rating 7.5A, 240V.
(b) Identify the device used to lower the voltage and describe how it works.
(c) Establish the cost of unit of electricity if in a month, your uncle pays a total of Ugy. 121,050, for which the fridge operates four hours every day for 15 days, each bulb works for twelve hours each day for 30 days and the phones spend one and a half hours charging every day for 30 days.
(d) Identify any two common house wiring components and explain to the villagers how each component is important.

Item 6 [MASAKA]

A certain landlady owns two roomed well-ventilated houses in an area that has extremely high temperatures. She has decided to install fans rated 240 V, 800 W powered by a DC motor to help cool the rooms. She is worried about the cost of powering the fans for 6 hours daily, and wanted a detailed explanation of how they work before installing them.

After installing the fans, she noticed the fans were not cooling the rooms well enough due to their low speed, and they had been connected in a way that they could only work together which would increase the electricity bill.

Hint: 1 KW of electricity costs U sh.950

Use your knowledge of Physics to;

a) To determine the daily cost of powering each of the fans.

b) Help the landlady understand how the fan works.

c) Explain to the landlady how the issues connected to the fans can be solved.

Item 7 [WAKISSHA]

A business man who operates a laundry business in one of the main trading centres in a district was charged Shs. 300,168 and is challenged with high electricity bill every month and feels cheated. He claims to operate two flat irons rated 1200 W each, for 2 hours per day, washing machine rated 2,200 W for 3 hours per day, and Three bulbs rated 100 W each for 10 hours per day. He thinks the rotating device in the washing machine over consumes power and therefore wishes to know how it operates.

Task

As a learner of physics;

(a) help the business man to know if he was being cheated or not

Hint: the unit cost of electricity is Shs.750

(b) Explain to the businessman to know how rotating device in the washing machine works.

(c) Advise on the possible means how these costs can be minimized.

ITEM 7 [JJEB]

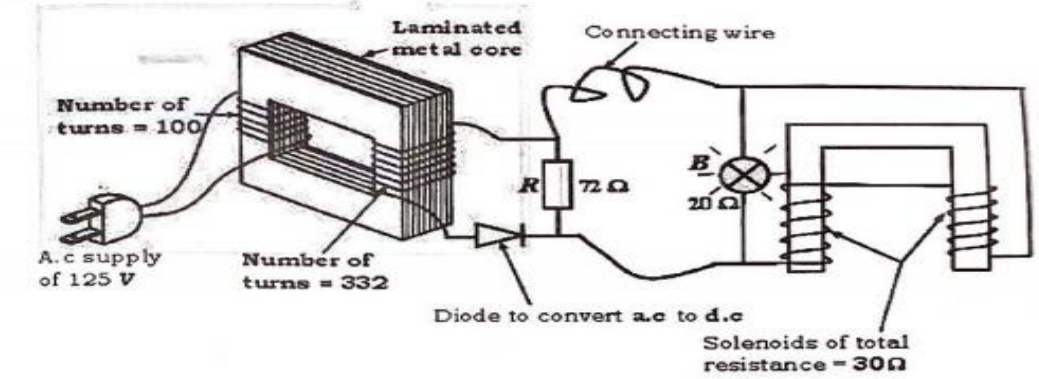
Metal scrap factories play an important role in recycling metal wastes that would otherwise cause pollution in the environment.

A technical director from one of the scrap factories is facing a critical issue involving delays in packing scrap metal into the crushing machine. He is searching for a person to design system capable of lifting a large volume of scrap from the off-load site to the crushing machine, and explain to the director how it works before he pays for it. In response, a physics club from one of the secondary schools in Uganda has organized an exhibition showcasing an identical system designed by a group of students, which they believe meets the director's objectives.

The technical specifications of the project report highlight that the system functions well only when the current flowing through it is not less than **14.2A** supplied from a direct current **(d.c.)** source, rather than an alternating current **(a.c.)** source directly.

The technical director wishes to know whether this assembled system will work reliably for his factory scale, why a **d.c** source is required instead of an **a.c** source.

Support diagram



The system designed by the students. R is a protective resistor and B is a monitoring bulb.

The director also wants to know, in case the lifting mechanism is weak, what modifications can be made to strengthen the system to do the work in the shortest time possible.

Furthermore, workers in the factory have complained of overheating of the wires when the machine is running and require an investigation into the origin of this phenomenon.
You have been approached as part of the group carrying out the system project.

Task:

(a) Read the scenario and help to:

(i) Address the technical director's concerns.

(ii) Help the workers understand the origin of the phenomenon happening on the connection wires attached to the machine

(b) Explain any other **two** daily life applications of the scientific concepts used in the system setup described in the scenario.

ITEM 7 [KAMSSA]

An engineer was assigned to supervise the installation of electrical power at a newly established field that would operate for seven days. The field had no electrical wiring, but it was located near a power distribution line supplied with electricity generated from a hydroelectric power station in Jinja. The engineer planned to connect the field to the main supply and determine the cost of the electrical energy that would be consumed during the operation period.
The field personnel would use the following electrical appliances:

Appliance	**Rating**	**Duration of Use per Day**
4 LED bulbs	240V, 10W each	6 hours
Electric kettle	240V, 2.0kW	15 minutes
Phone charging station	240V, 120W	4 hours
Radio receiver	240V, 60W	5 hours
Laptop computer	240V, 90W	3 hours
Electric iron	240V, 1.2kW	20 minutes

The engineer learned that the electricity supplied to the camp was generated by an A.C. generator at a hydroelectric power station. During a technical briefing, the engineer was asked to explain how the A.C generator works, how power is transmitted and how the electricity bill for the camp would be determined for seven days.

Support

The electrical energy costs UGX 1,000 per kWh

Task

As a physics learner, respond to the requests from the engineer.

ITEM 7[UMTA]

A commercial laundry business operates a heavy-duty washing machine powered by a **240 V** DC motor. During a severe rainy season, a sudden lightning power surge causes an excessive current to flow through the motor, damaging its internal windings and halting business operations.

In order to prevent this damage from happening again, an electrician installs a safety protection circuit. This circuit consists of a **3 A** fuse connected in series with the **80 Ω DC** motor as shown in **Fig.3** below. To help the operator monitor the system, a light-emitting diode (LED) connected in series with an **18 kΩ** resistor is placed across the fuse to show whether the fuse is intact or has blown.

As the heavy rains continue weeks later, another lightning strike causes the main supply voltage to shoot up from **240 V to 300 V**. The laundry owner now needs to determine whether this newly installed fuse will successfully blow to protect the motor, and whether the LED indicator will correctly light up to show the condition of the fuse.

Hint: *The LED operates optimally at a current between **10 mA** and **20 mA***.

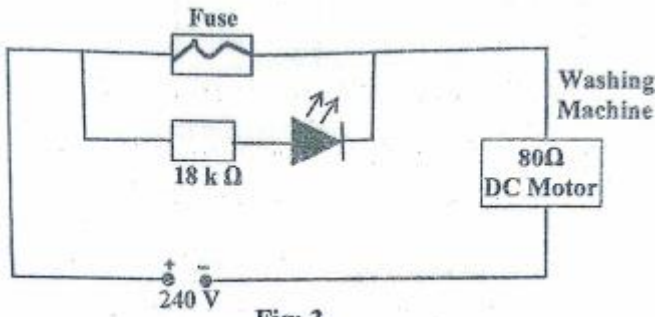


Fig: 3.

Task:

(a) Explain how the fuse, resistor and **LED** work together to protect and monitor the washing machine.

(b) Determine the current in the **LED** branch:

(i) before the fuse blows at **240 V**;

(ii) after the fuse has blown when the supply rises to **300** V.

Hence, assess whether the **LED** provides a reliable indication that the fuse has operated.

(c) Explain how the **DC** motor operates and state two factors that affect the turning effect produced.

(d) State precautions that should be taken to protect electrical appliances from damage during thunderstorms.

ITEM 7 [UTEC]

In an effort to connect the Karamoja region to the national electricity grid from the River Nile, local residents were told to clear and vacate a stretch of land which was a minimum of 10 feet

from the grid path. The transmission line is expected to carry a voltage of around 40 kV, which would later be stepped down to 14 kV using a transformer with 400 turns in its primary coil. The locals disagreed, claiming that all that land could not be left unutilized just because of a mere electric grid pathway. The officials then requested the Chairman L.C.1 of one of the villages to call for an awareness meeting. The chairman used a bicycle to ride around the village mobilizing residents. He tried turning on the headlight, but it did not work; he was told by a mechanic that the dynamo had internal technical issues, which was the source of light. The Chairman wondered greatly how the bicycle dynamo works to produce light without any dry cell battery.

Task:

a) Help convince the locals to appreciate the safety need to leave the demarcated grid space unutilized.
b) Kindly clear the chairman's wonder by explaining how a bicycle dynamo generates electricity.
c) With reference to calculations, advise on the exact specifications (number of turns in the secondary coil) of the transformer required to step down the voltage as specified.
d) Using a well-labeled diagram, explain to the locals the path of grid transmission from the River Nile power station to their region.
e) Help locals understand why electric power is transmitted along the grid at such a very high voltage.
f) Discuss more applications of the device used to step down or step-up voltage in everyday technology.

ITEM 7 [WENSSEC]

"Out of nowhere, we heard like a thunderstorm" a witness said. "It was extremely loud." "And all of a sudden, the building housing the offices was on fire burning." "I really don't understand what happened." After the incident, the witness donated a faulty device rated 108 W, 240 V for use while addressing the public during community mobilization for building materials in an attempt to rebuild the burnt building. It was later discovered that the device needed a small size resistor that could allow a current not exceeding 13 A to function well again. Two rods of iron and copper were also donated to help in safe guarding the building after reconstruction.

Task.

Help;

(a) The witness with all relevant information so that he understands what really happened in the community:

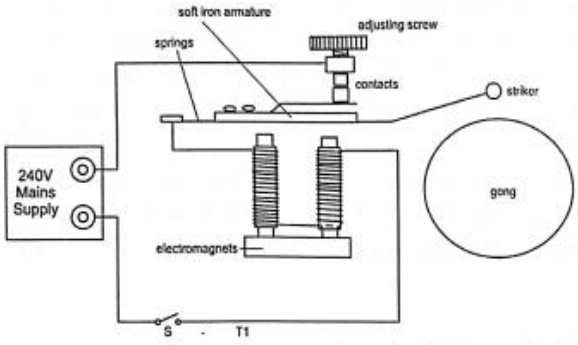
(b) With the value of resistance required to make the device function again.

(c) Understand how the device achieves the intended purpose

(d) Understand how the new building can be safe guarded to prevent future occurrence of the same incident.

ITEM 7 [MASAKA]

A man who owns a house very far from the gate could not hear whenever they knocked at his gate. He was advised to install an electric bell, which his son decided to make for him if he was given a design. He was shown the illustration below.



As the son was making the electric bell, he made the following changes;

- He replaced the soft iron armature with a piece of steel.
- He used a coil of 50 ohms instead of another of 24 ohms.

The man demanded his son to explain to him how this device would solve the problem, how the changes his son made would affect its operation. He also demanded to know how to protect the device from power surges like his home radio had recently done.

Use your knowledge of Physics;

a) To help the man understand how the device will solve his problem.

b) To explain how the changes made by the son will affect the operation of the device.

c) Suggest how the device can be protected.

PHYSICS DEPARTMENT @KSCK

“KEEP PRACTICING”

END